

Question	Answer	Marks
5(a)(i)	cross labelled N marked at the intersection of the solid and dashed lines or at X or Y and cross labelled A marked at a peak or a trough	B1
5(a)(ii)	(XY is 1.5 wavelengths, so) wavelength = $0.48 \times (2 / 3) = 0.32 \text{ m}$ or (wavelength is twice node–node distance so) $\lambda / 2 = 0.48 / 3 = 0.16 \text{ m}$ and wavelength = $0.16 \times 2 = 0.32 \text{ m}$	B1
5(a)(iii)	$v = f\lambda$	C1
	frequency = $1400 / 0.32$ = 4400 Hz	A1
5(b)(i)	$f_o = f_s v / (v - v_s)$ $f_o = (780 \times 320) / (320 - 39)$	C1
	maximum frequency = 890 Hz	A1
5(b)(ii)	line showing f varying both above and below a mean frequency and returning to the original start value of f	B1
	a single cycle of a smoothly oscillating curve of correct phase (starting at mean position, falling to a trough, then rising to a peak, ending at mean position)	B1

9702/23

Cambridge International AS & A Level – Mark Scheme

October/November 2024

PUBLISHED

Question	Answer	Marks